

Leetcode 81 :-

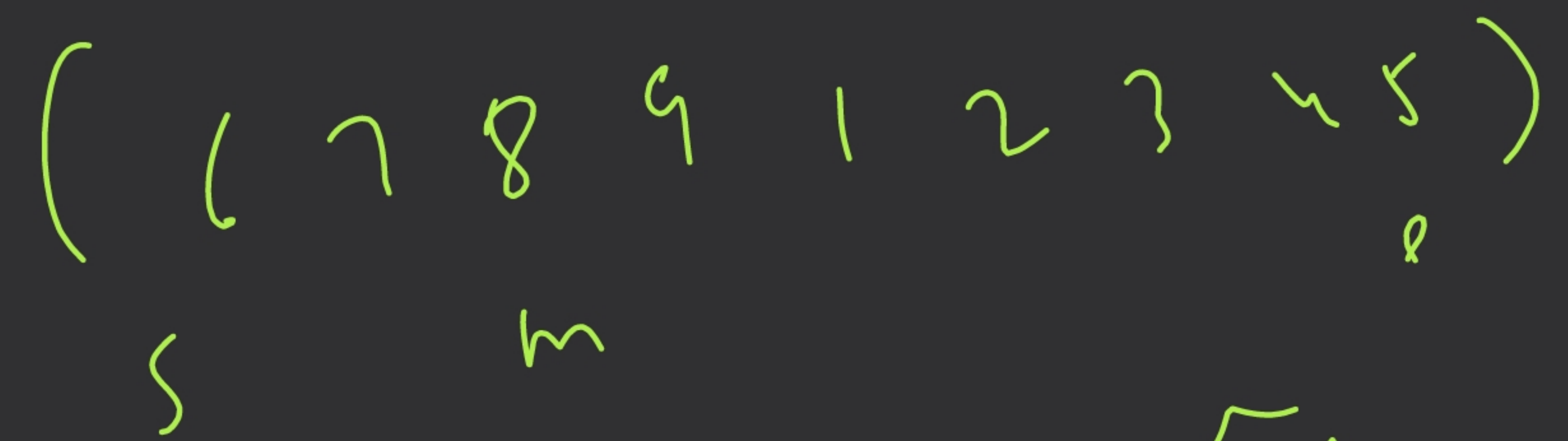
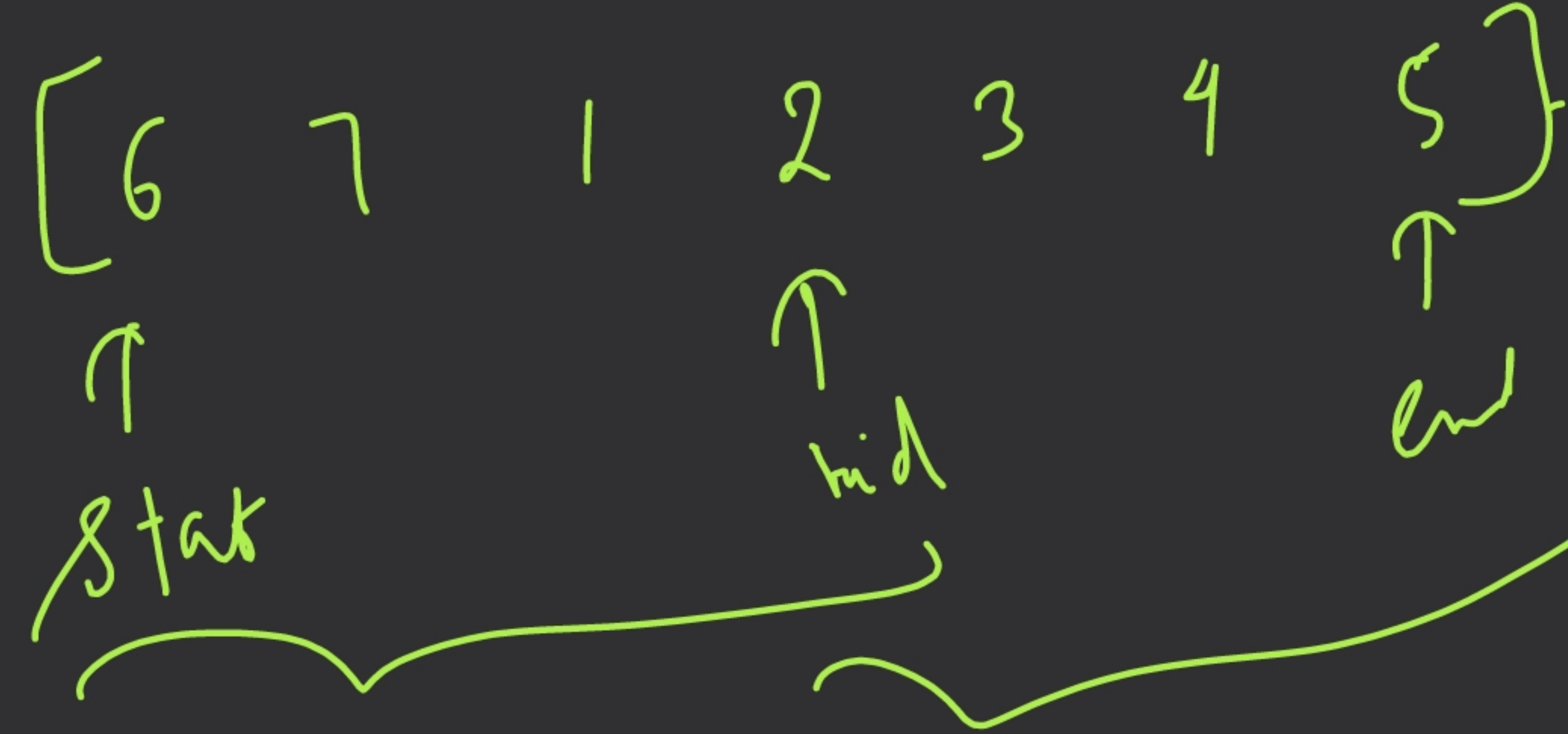
Search in Rotated Sorted Array - II -

(Ist & 2nd part)

2

$[8 \quad 10 \quad 12 \quad 1 \quad 2 \quad 3]$
 $\uparrow \quad \quad \uparrow \quad \quad \quad \quad \uparrow$
 $low \quad \quad \quad mid \quad \quad \quad \quad \quad \quad high$

$$(8 < 10) \checkmark$$



left +
start

$arr(start) < = arr(mid)$
 target right
 target outside

$s = mid + 1$

$e = mid - 1$

right

$arr(start) > arr(mid)$

target slow
 mid & end

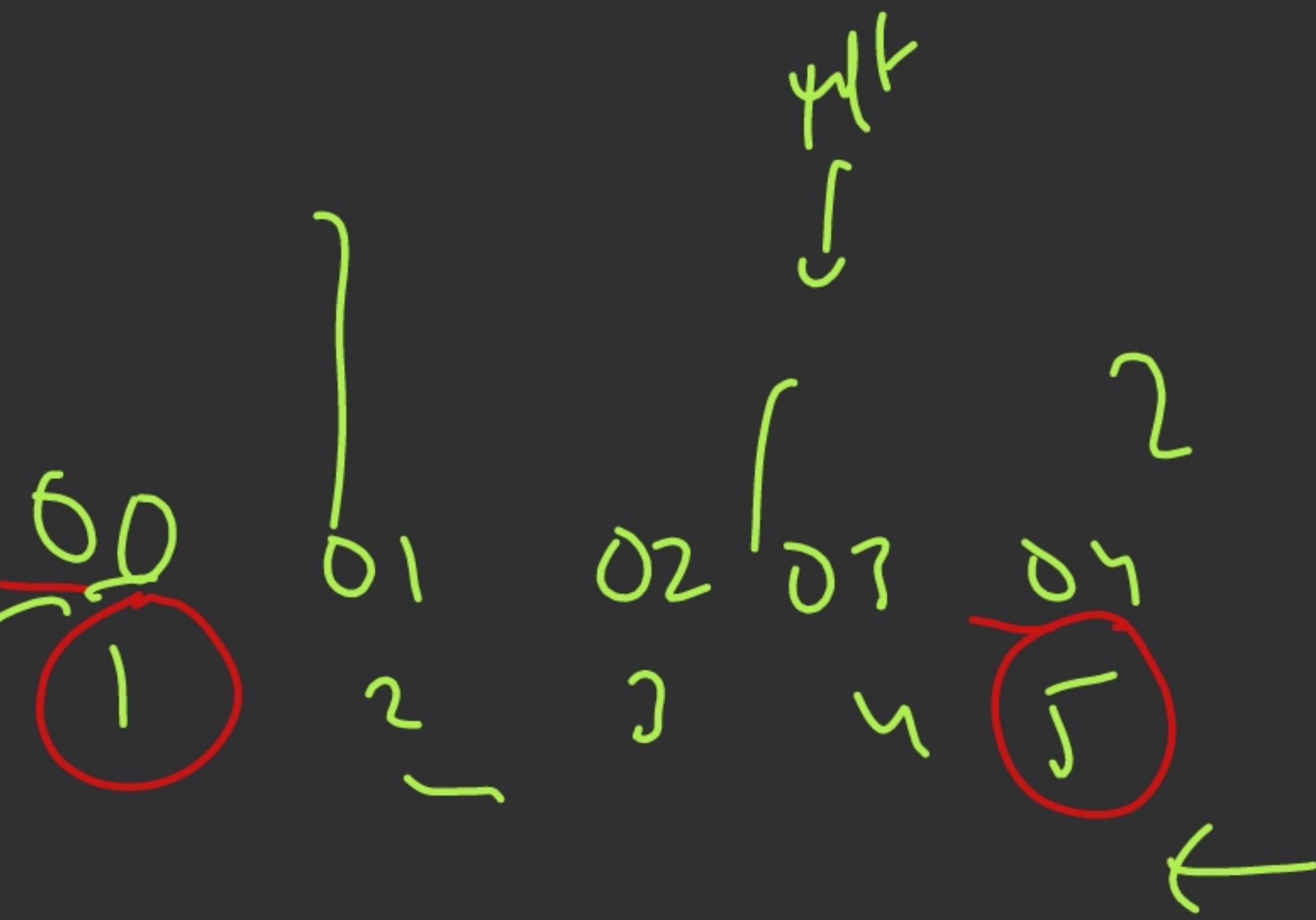
$e = mid - 1$

target
outside

$s = mid + 1$

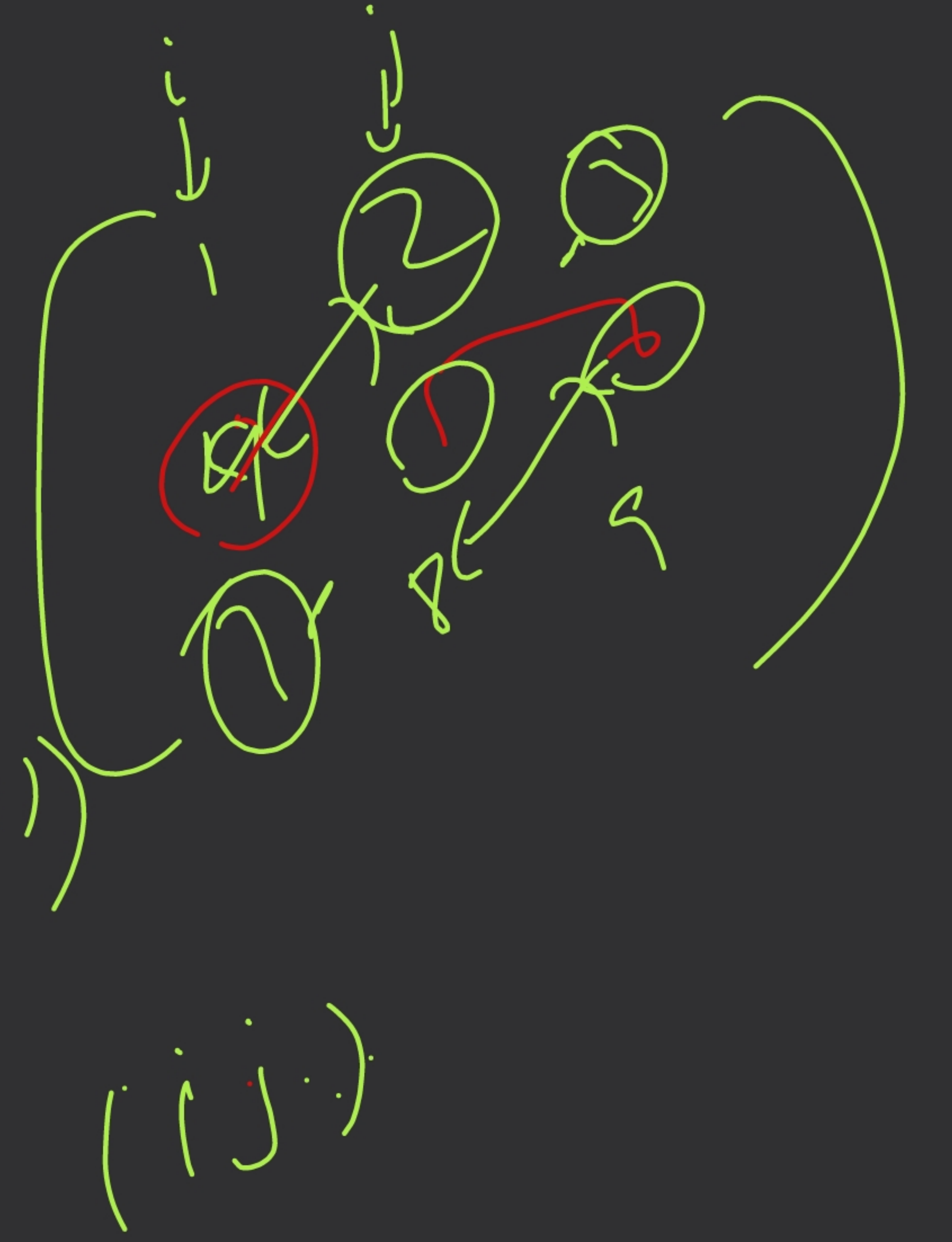
// Transpon

```
for (i = 0; i < n; i++) {  
    for (j = i + 1; j < n; j++) {  
        swap(arr[i][j], arr[j][i])  
    }  
}
```



// Rotation

```
for (i = 0; i < n; i++) {  
    int left = 0, right = m - 1;  
    while (left < right) {  
        swap(arr[i][left], arr[i][right])  
        left++, right--  
    }  
}
```



Matrix multiplication

Col of
1st matrix

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \quad 3 \times 2$$

= row of
2nd matrix

$$(\underline{m} \times \underline{n}) \times (\underline{n} \times \underline{p})$$

(new matrix of $m \times p$)

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \quad 2 \times 3$$

$$\begin{bmatrix} 1 \times 1 + 2 \times 4 & 1 \times 2 + 2 \times 5 \\ 3 \times 1 + 4 \times 4 & 3 \times 2 + 4 \times 5 \\ 5 \times 1 + 6 \times 4 & 5 \times 2 + 6 \times 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 \times 3 + 2 \times 6 \\ 3 \times 3 + 4 \times 6 \\ 5 \times 3 + 6 \times 6 \end{bmatrix} \quad 3 \times 3$$

Ques.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \quad 2 \times 3$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \quad 3 \times 2$$